

Midterm Activity #1 - Variables

KARIM, Denisse Jane F.

BSIT 2-3

NO. _____

DATE _____

COMP 009 - OOP

Denisse F.

Part A - Variable Identification Table

Variable Name	Data Type	Variable Type	stored in
owner	String	Instance	Heap
balance	double	Instance	Heap
acc1	Account	Local	stack
acc2	Account	Local	stack
acc3	Account	Local	stack
args	String[]	Local	stack

Part B: Explanation

Statement	What happens?	Memory Impact
Account acc1 = new Account("Carlos", 8000);	Creates an Account object with owner Carlos and balance of 8000	New object allocated in Heap, reference acc1 stored in stack
Account acc2 = new Account("Diana", 7000);	Creates an Account object with owner Diana and balance of 7000	New object allocated in Heap, reference acc2 stored in stack
Account acc3 = acc1	acc3 points to the same object as acc1	Reference acc3 stored in stack, both acc1, acc3 point to same Heap object
acc2.balance = 8000;	Updates the balance of the object referenced by acc2	Value inside Heap object is updated
acc3.owner = "Miguel";	Changes the owner of object referenced by acc3. since acc3 = acc1, it changes acc1's object too.	Field inside the same Heap object updated

part C - Output

Initial :

acc1 → ("Carlos", 5000)

acc2 → ("Diana", 7000)

acc3 → acc1

After Modification :

acc2.balance = 8000

acc3.owner = "Miguel"

Program Output

Miguel	5000.0
Diana	8000.0
Miguel	5000.0

//

→ prints acc1

→ prints acc2

→ prints acc3